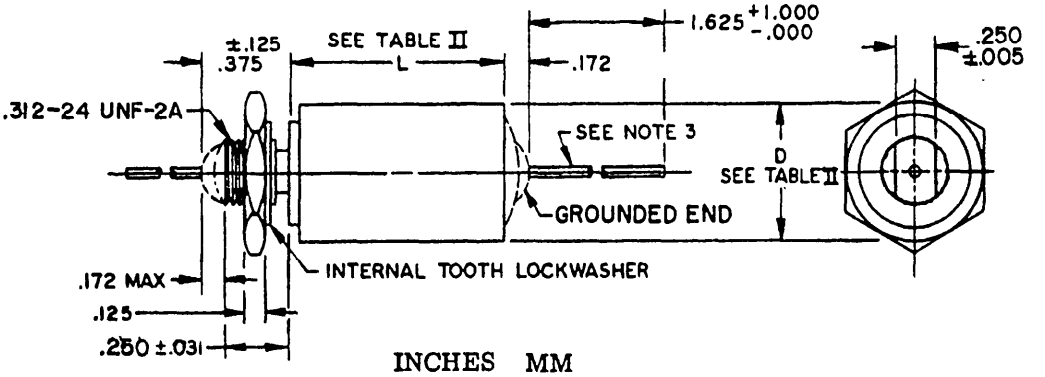
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL, THREADED-STUD RETAINER, TUBULAR (UNINSULATED), (HERMETICALLY SEALED IN METAL CASES), STYLE CQ11

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-19978.

INACTIVE FOR NEW DESIGN AFTER
30 August 1974



INCHES	MM
.001	.03
.004	.10
.005	.13
.031	.79
.032	.81
.125	3.18
.172	4.37
.250	6.35
.312	7.92
.375	9.53
1.000	25.40
1.625	41.28

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Leads shall be of tinned solid wire, .032 +.004, -.001 (No. 20 AWG).
4. Mounting nuts and lockwashers shall be supplied.

FIGURE 1. Style CQ11 capacitors.

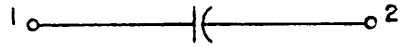
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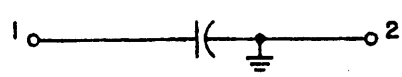
REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1 and table II.
Case material - Nonmagnetic (screwneck and end seal may be of magnetic material).
Dielectric material - Paper and polyethylene terephthalate (characteristic K).
Terminals - Axial-wire lead (see figure 1).

Circuit diagram - 1



3



Rated voltage - See table II.
Rated temperature - -65° to +125°C.
Capacitance (Cap.) (nom) - See table II.
Capacitance tolerance - ±10 percent (symbol K).
Vibration grade - 3.
Dissipation factor (DF) (max) - 1.0 percent.

Seal: Method 112 of MIL-STD-202, test condition letter - A.

Barometric pressure: In accordance with MIL-C-19978.
Test points:
Circuit diagram 1 - Between terminals and case.
Circuit diagram 3 - Between ungrounded terminal and case.

Insulation resistance (IR):
Terminal to terminal - See table I.

TABLE I. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
	At 25°C
0 to 0.6 microfarad	25,000 megohms
Greater than 0.6 microfarad	15,000 megohm-microfarads 1/
	At 125°C
0 to 0.08 microfarad	250 megohms
Greater than 0.08 microfarad	20 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Terminals to case - Greater than 10,000 megohms.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:
Direction and duration of motion - 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: In accordance with MIL-C-19978.

Solderability: In accordance with MIL-C-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter - D.

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TABLE II. Style CQ11 capacitors.

Part number 1/	DC voltage rating	Capacitance	Case dimensions (in inches)		
			D +.010 -.020	L ±.015	
				Circuit 1	Circuit 2
	Volts	µF			
CQ11A-KB333K3	100	0.033	0.400	0.771	0.708
CQ11A-KB154K3	100	0.15	0.400	0.896	0.833
CQ11A-KB224K3	100	0.22	0.400	1.146	1.083
CQ11A-KB394K3	100	0.39	0.562	1.156	1.094
CQ11A-KB474K3	100	0.47	0.562	1.156	1.094
CQ11A-KB684K3	100	0.68	0.562	1.406	1.344
CQ11A-KB105K3	100	1.0	0.562	1.906	1.844
CQ11A-KC683K3	200	0.068	0.400	0.771	0.708
CQ11A-KC104K3	200	0.10	0.400	0.896	0.833
CQ11A-KC154K3	200	0.15	0.400	1.146	1.083
CQ11A-KC224K3	200	0.22	0.400	1.396	1.333
CQ11A-KC334K3	200	0.33	0.562	1.156	1.094
CQ11A-KC474K3	200	0.47	0.562	1.406	1.344
CQ11A-KC684K3	200	0.68	0.562	1.906	1.844
CQ11A-KC105K3	200	1.0	0.670	1.906	1.844
CQ11A-KE103K3	400	0.010	0.400	0.771	0.708
CQ11A-KE393K3	400	0.039	0.400	0.896	0.833
CQ11A-KE473K3	400	0.047	0.400	0.896	0.833
CQ11A-KE683K3	400	0.068	0.400	1.146	1.083
CQ11A-KE104K3	400	0.10	0.400	1.396	1.333
CQ11A-KE154K3	400	0.15	0.562	1.156	1.094
CQ11A-KE224K3	400	0.22	0.562	1.406	1.344
CQ11A-KE274K3	400	0.27	0.562	1.906	1.844
CQ11A-KE334K3	400	0.33	0.562	1.906	1.844
CQ11A-KE474K3	400	0.47	0.670	1.906	1.844
CQ11A-KE684K3	400	0.68	0.750	2.156	2.094
CQ11A-KE105K3	400	1.0	1.000	1.906	1.844
CQ11A-KF102K3	600	0.001	0.400	0.771	0.708
CQ11A-KF152K3	600	0.0015	0.400	0.771	0.708
CQ11A-KF222K3	600	0.0022	0.400	0.771	0.708
CQ11A-KF332K3	600	0.0033	0.400	0.771	0.708
CQ11A-KF472K3	600	0.0047	0.400	0.771	0.708
CQ11A-KF682K3	600	0.0068	0.400	0.771	0.708
CQ11A-KF103K3	600	0.010	0.400	0.771	0.708
CQ11A-KF153K3	600	0.015	0.400	0.771	0.708
CQ11A-KF223K3	600	0.022	0.400	0.771	0.708
CQ11A-KF333K3	600	0.033	0.400	0.896	0.833
CQ11A-KF473K3	600	0.047	0.400	1.146	1.083
CQ11A-KF683K3	600	0.068	0.400	1.396	1.333
CQ11A-KF104K3	600	0.10	0.562	1.156	1.094
CQ11A-KF154K3	600	0.15	0.562	1.406	1.344
CQ11A-KF184K3	600	0.18	0.562	1.906	1.844
CQ11A-KF224K3	600	0.22	0.562	1.906	1.844
CQ11A-KF334K3	600	0.33	0.670	1.906	1.844
CQ11A-KF474K3	600	0.47	0.750	2.156	2.094
CQ11A-KF684K3	600	0.68	1.000	1.906	1.844

1/ Complete part number shall include an additional symbol to indicate circuit.

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TABLE III. Millimeter equivalents of decimal inches.

Inches	mm	Inches	mm	Inches	mm
0.010	0.25	0.771	19.58	1.333	33.86
0.015	0.38	0.833	21.16	1.344	34.14
0.020	0.51	0.896	22.76	1.396	35.46
0.400	10.16	1.000	25.40	1.406	35.71
0.562	14.27	1.083	27.51	1.844	46.84
0.670	17.02	1.094	27.79	1.906	48.41
0.708	17.98	1.146	29.11	2.094	53.19
0.750	19.05	1.156	29.36	2.156	54.76

Stability at low and high temperature:
Low temperature:
Test temperature - $-65^{\circ}\ ^{+0^{\circ}}_{-5^{\circ}}$C.
Capacitance change (max) - -10 percent.
High temperature:
Test temperature - $+125^{\circ}\ ^{+5^{\circ}}_{-0^{\circ}}$C.
Capacitance change (max) - +10 percent.

Life:
Capacitance change (max) - ±5 percent.

Resistance to soldering heat: In accordance with MIL-C-19978.

NOTE: Style CP11 capacitors formerly covered in MIL-C-25/1K are substitutable for capacitors covered in this specification sheet as follows:

From MIL-C-25/1K (CP11)	To MIL-C-19978/21 (CQ11)	From MIL-C-25/1K (CP11)	To MIL-C-19978/21 (CQ11)
CP11A-KB333K-	CQ11A-KB333K3	CP11A-KE334K-	CQ11A-KE334K3
CP11A-KB154K-	CQ11A-KB154K3	CP11A-KE474K-	CQ11A-KE474K3
CP11A-KB224K-	CQ11A-KB224K3	CP11A-KE684K-	CQ11A-KE684K3
CP11A-KB394K-	CQ11A-KB394K3	CP11A-KE105K-	CQ11A-KE105K3
CP11A-KB474K-	CQ11A-KB474K3	CP11A-KF102K-	CQ11A-KF102K3
CP11A-KB684K-	CQ11A-KB684K3	CP11A-KF152K-	CQ11A-KF152K3
CP11A-KB105K-	CQ11A-KB105K3	CP11A-KF222K-	CQ11A-KF222K3
CP11A-KC683K-	CQ11A-KC683K3	CP11A-KF332K-	CQ11A-KF332K3
CP11A-KC104K-	CQ11A-KC104K3	CP11A-KF472K-	CQ11A-KF472K3
CP11A-KC154K-	CQ11A-KC154K3	CP11A-KF682K-	CQ11A-KF682K3
CP11A-KC224K-	CQ11A-KC224K3	CP11A-KF103K-	CQ11A-KF103K3
CP11A-KC334K-	CQ11A-KC334K3	CP11A-KF153K-	CQ11A-KF153K3
CP11A-KC474K-	CQ11A-KC474K3	CP11A-KF223K-	CQ11A-KF223K3
CP11A-KC684K-	CQ11A-KC684K3	CP11A-KF333K-	CQ11A-KF333K3
CP11A-KC105K-	CQ11A-KC105K3	CP11A-KF473K-	CQ11A-KF473K3
CP11A-KE103K-	CQ11A-KE103K3	CP11A-KF683K-	CQ11A-KF683K3
CP11A-KE393K-	CQ11A-KE393K3	CP11A-KF104K-	CQ11A-KF104K3
CP11A-KE473K-	CQ11A-KE473K3	CP11A-KF154K-	CQ11A-KF154K3
CP11A-KE683K-	CQ11A-KE683K3	CP11A-KF184K-	CQ11A-KF184K3
CP11A-KE104K-	CQ11A-KE104K3	CP11A-KF224K-	CQ11A-KF224K3
CP11A-KE154K-	CQ11A-KE154K3	CP11A-KF334K-	CQ11A-KF334K3
CP11A-KE224K-	CQ11A-KE224K3	CP11A-KF474K-	CQ11A-KF474K3
CP11A-KE274K-	CQ11A-KE274K3	CP11A-KF684K-	CQ11A-KF684K3

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